

MULTI-AGENT DATA ANALYSIS AND INSIGHT GENERATION PLATFORM FOR FINANCIAL AND SALES DATA

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18 July, 2026

Presentation Outline

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- Objectives
- LLM Architecture
- Methodology
- Current Progress
- Mid-Progress Timeline
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Introduction

- The project is being implemented as a state-based data analysis pipeline for sales and financial CSV datasets.
- Four pipeline stages are currently connected: structural profiling, semantic tagging, preprocessing, and statistical analysis.
- The system transforms raw records into a cleaned dataset, a preprocessing audit trail, quality metrics, statistics, and chart files.
- Mathematical operations are performed with Python libraries; the language model is used for semantic interpretation and processing rules.
- The current implementation provides a working backend foundation for the planned user-facing platform.

Objectives

- Develop an autonomous multi-agent system that converts raw sales and financial data into reproducible analytical outputs.
- Implement semantic, context-aware preprocessing instead of applying the same cleaning rule to every column.
- Evaluate the pipeline using representative datasets, focused unit tests, and generated diagnostics.
- Establish a reliable backend foundation before completing report generation and the user interface.

LLM Architecture – Llama Transformer

- The model used in the pipeline is Llama, a decoder-only transformer language model.
- It processes the input context through repeated self-attention and feed-forward blocks.
- The attention layers help the model relate each token to earlier tokens in the prompt.
- This architecture makes it suitable for semantic tagging, schema inference, and structured reasoning.

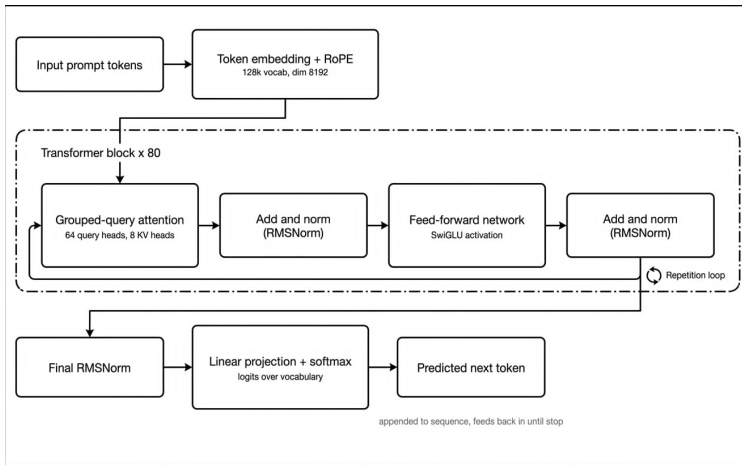


Figure 1: Llama Transformer architecture

Methodology – Current 4-Stage Pipeline

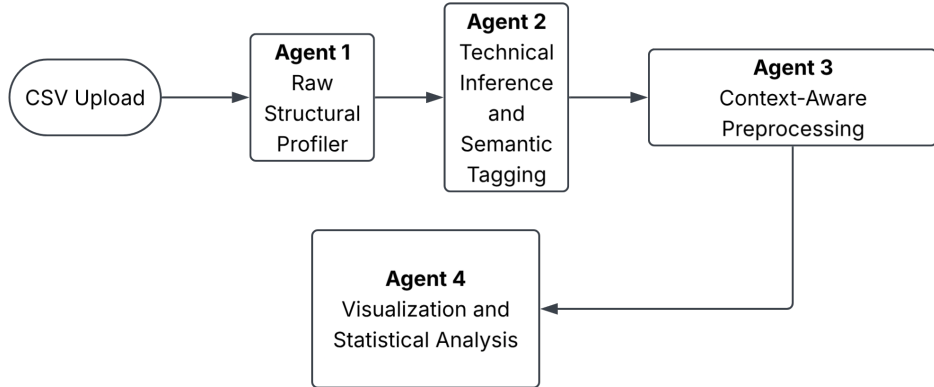


Figure 2: Current 4-stage pipeline

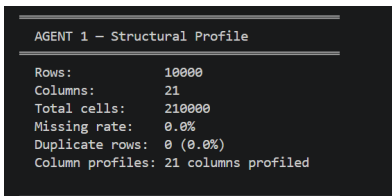
Current Progress

Implementation of the 4-Stage Pipeline

- Agent 1: Raw Structural Profiler
- Agent 2: Technical Inference & Semantic Tagging
- Agent 3: Context-Aware Preprocessing
- Agent 4: Visualization & Statistical Analysis

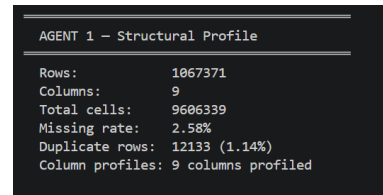
Agent 1: Raw Structural Profiler

- Reads the input CSV and records shape, dtypes, missingness, uniqueness, and duplicates.
- Produces a raw structural profile without changing the source data.
- Caches the DataFrame for the downstream stages.

A terminal window showing the output of the 'AGENT 1 - Structural Profile' command. The output is a text-based summary of the data's structure.

AGENT 1 - Structural Profile	
Rows:	10000
Columns:	21
Total cells:	210000
Missing rate:	0.0%
Duplicate rows:	0 (0.0%)
Column profiles:	21 columns profiled

Figure 3: Agent 1 profiling view

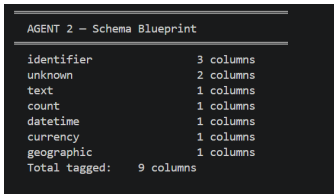
A terminal window showing the output of the 'AGENT 1 - Structural Profile' command. The output is a text-based summary of the data's structure.

AGENT 1 - Structural Profile	
Rows:	1067371
Columns:	9
Total cells:	9606339
Missing rate:	2.58%
Duplicate rows:	12133 (1.14%)
Column profiles:	9 columns profiled

Figure 4: Agent 1 output summary

Agent 2: Technical Inference & Semantic Tagging

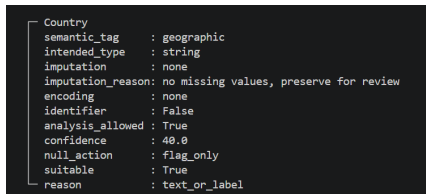
- Sniffs numeric, datetime, boolean, and string types using local heuristics.
- Uses Groq-based semantic tagging when available and falls back to deterministic rules when it is not.
- Creates a per-column blueprint containing type, semantic tag, scaling, imputation, and analysis policies.
- Passes the blueprint to preprocessing instead of hard-coding one rule for every column.



AGENT 2 - Schema Blueprint

identifier	3 columns
unknown	2 columns
text	1 columns
count	1 columns
datetime	1 columns
currency	1 columns
geographic	1 columns
Total tagged:	9 columns

Figure 5: Agent 2 semantic blueprint



```
[ Country
  semantic_tag : geographic
  intended_type : string
  imputation : none
  imputation_reason: no missing values, preserve for review
  encoding : none
  identifier : False
  analysis_allowed : True
  confidence : 40.0
  null_action : flag_only
  suitable : True
  reason : text_or_label
```

Figure 6: Agent 2 fallback tagging output

Agent 3: Context-Aware Preprocessing

- Coerces columns according to the semantic blueprint and cleans international currency formats.
- Standardizes text and null variants while preserving identifiers.
- Removes duplicates, applies configured imputation, and clips permitted outliers with IQR bounds.
- Extracts date features and derives metrics such as profit, margin, and revenue per unit.
- Records scaling parameters, an audit log, and a 0–100 data quality score.

```
AGENT 3 – Preprocessing

Quality score: 98.47
Completeness: None%
Duplicates removed: None
Cleaned shape: 812368 rows x 21 cols
Preprocessing steps logged: 21
Cleaned CSV: outputs\cleaned_data.csv
```

Figure 7: Agent 3 preprocessing preview

```
[Agent 3] Preprocessing: profile=strict domain=finance_sales input=1067371x9
[Agent 3] Completed: rows=1067371->812368 cols=9->21 quality=98.47/100 raw_missing=2.58% remaining_nulls
~0.0%
[Agent 3] Cleaning summary: deduped=12133, scaled=0, added_cols=12, removed_rows=255803
[Agent 3] Note: the cleaned dataset is normalized and exported to outputs/cleaned_data.csv; if the row c
ount stays the same, that means there were no exact duplicates or rows to drop.
```

Figure 8: Agent 3 audit output

Agent 4: Visualization & Statistical Analysis

- Calculates descriptive statistics, correlations, distributions, growth, and seasonality.
- Detects anomalies using a configurable z-score threshold.
- Computes regression trends and category rankings when the dataset supports them.
- Saves reproducible PNG charts and structured statistics to the pipeline state.

```
[Agent 4] Starting analysis: 812368 rows x 21 cols  
[Agent 4] Step 1 - Descriptive stats: 10 columns  
[Agent 4] Step 2 - Correlation done, strong pairs: 0  
[Agent 4] Step 3 - Growth rates skipped (no revenue/time axis)  
[Agent 4] Step 4 - Rankings skipped (no revenue/category pairing)  
[Agent 4] Step 5 - Seasonality skipped (no time axis)  
[Agent 4] Step 6 - Anomalies: 23915 unique rows (2.94%) across 3 columns  
[Agent 4] Step 7 - Distributions done (0 charts)  
[Agent 4] Step 8 - Regression skipped (no time axis)  
[Agent 4] Step 9 - Distribution charts done (1 charts)  
[Agent 4] Step 10 - Derived metrics skipped (no derived metrics)  
[Agent 4] Done - 1 charts saved to outputs/charts/
```

Figure 9: Agent 4 statistical output

```
=====
```

```
AGENT 4 - Statistical Analysis
```

```
=====
```

```
Descriptive stats columns: 10  
Strong correlation pairs: 0  
Anomalous rows: 23915 (2.94%)  
Charts saved: 1  
    outputs/charts\boxplot_numeric_cols.png
```

```
=====
```

Figure 10: Agent 4 chart output

Mid-Progress Timeline

- **Completed:** project architecture, GraphState, Agents 1–3, conditional pipeline routing, preprocessing audit trail, quality scoring, and focused tests.
- **Current:** Agent 4 statistical analysis and chart generation, verified on the Amazon sales dataset.
- **Next:** connect statistics to evidence-backed insight generation.
- **Final phase:** upload workflow, frontend presentation, broader datasets, and end-to-end evaluation.

Remaining Work

Work package	Target outcome
Validation guardrail	Cross-check statistics, charts, and evidence
Insight generation	Produce traceable natural-language findings
Interface	Upload data and display progress/results
Evaluation	Benchmark accuracy, robustness, and run-time

Conclusion

- **Backend foundation achieved:** A working LangGraph pipeline now carries a CSV dataset through four specialized stages.
- **Core transformation achieved:** Profiling, semantic tagging, context-aware preprocessing, quality scoring, and statistical chart generation are operational.
- **Evidence available:** The verified run produces cleaned data, diagnostics, audit records, statistics, and chart artifacts.
- **Next objective:** complete validation, evidence-backed report generation, interface integration, and broader evaluation.

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